Image Grayscale Conversion

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1. Implementation Principle
2. Implementation Effect
3. Implementation Code

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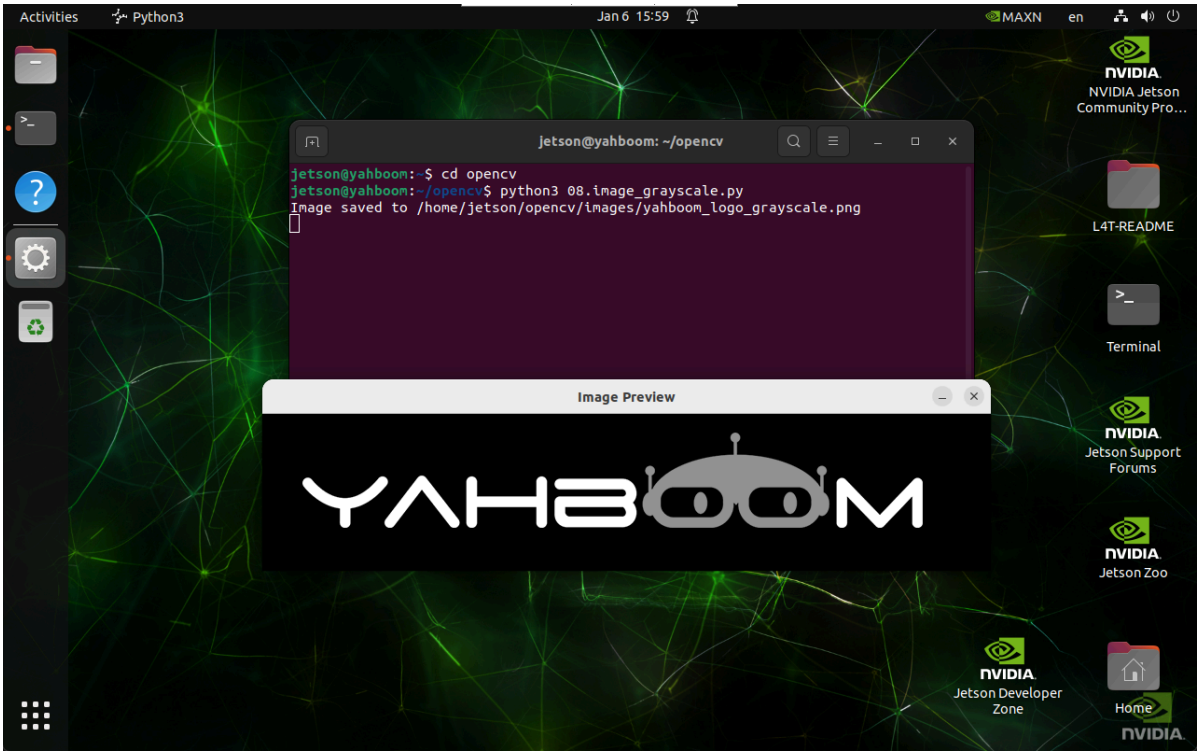
Use the `cv2.cvtColor()` function to convert the color space.

2. Implementation Effect

```
cd ~/opencv
```

```
python3 08.image_grayscale.py
```

> Note: click the image window and press any key to exit.



3. Implementation Code

```
import cv2

def grayscale_image(input_path, output_path):
    image = cv2.imread(input_path)
    if image is None:
        print("Error: Unable to open image file.")
        return
    gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    if cv2.imwrite(output_path, gray_image):
```

```
        print(f"Image saved to {output_path}")
        cv2.imshow('Image Preview', cv2.imread(output_path))
        cv2.waitKey(0)
        cv2.destroyAllWindows()
    else:
        print("Error: Unable to save image file.")

grayscale_image('/home/jetson/opencv/images/yahboom_logo.png', \
                '/home/jetson/opencv/images/yahboom_logo_grayscale.png')
```